



Edit Mode is: ON



BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN

Tests, Surveys, and Pools Tests

Test Canvas : SAE 01

Test Canvas: SAE 01

The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)

Create Question

Reuse Question

Upload Questions



Question Settings

Description Avail: 2025 Jan 14-2300 to Feb 02-2359
(W1 last class to W3 Sun midnight)
Normally till W1 Sun midnight
Max Attempts: 09
Grade: Highest Graded
Password: 154

Instructions

Total Questions 12

Total Points 12

Select: All None

Select by Type:

- Question Type -



Delete

Points

Update

Hide Question Details



Points: 1

1. True/False: BU5401-2024S2::SAE01::Q101::TF::
(95, ...



Question

In a linear programming problem, the constraints must be linear, but the objective function may be non-linear.

Answer

True

False



Points: 1

2. True/False: BU5401-2024S2::SAE01::Q201::TF::
(120,...



← OK

Question The term 'deterministic' means the model's answer(s) can be determined to be the same all the time.

Answer ☒ True
☐ False



Points: 1

3. True/False: BU5401-2024S2::SAE01::Q301::TF::
(144,...



Question In management problem solving, any solution that works is called an 'optimal solution'.

Answer ☐ True
☒ False



Points: 1

4. True/False: BU5401-2024S2::SAE01::Q401::TF::
(169,...



Question Linear programming model is an example of mathematical modeling.

Answer ☒ True
☐ False



Points: 1

5. True/False: BU5401-2024S2::SAE01::Q501::TF::
(194,...



Question A toy train layout designed to represent an actual railyard is an example of a mathematical model.

Answer ☐ True
☒ False



Points: 1

6. True/False: BU5401-2024S2::SAE01::Q601::TF::
(218,...



Question A feasible solution which satisfies all given constraints in a model is an optimal solution.

Answer	True
	☒ False



Points: 1

7. True/False: BU5401-2024S2::SAE01::Q701::TF::
(242,...



Question	An optimal solution is 'optimal' in the sense that the objective function value has reached the extreme.
----------	--

Answer	☒ True
	False



Points: 1

8. True/False: BU5401-2024S2::SAE01::Q801::TF::
(267,...



Question	Breakeven analysis is too simple to be useful in practice.
----------	--

Answer	True
	☒ False



Points: 1

9. True/False: BU5401-2024S2::SAE01::Q901::TF::
(292,...



Question	A company tries to minimize its total costs subject to producing decent-enough profits. The total-cost is a constraint here.
----------	--

Answer	True
	☒ False



Points: 1

10. Calculated Numeric:
BU5401-2024S2::SAE01::Q1001::NUM::(32...



Question You are given the LP Model as follows:

$$\text{Max } 12X + 13Y$$

such that

$$X + 33Y \leq 41 \quad (\text{C1})$$

$$45X + 20Y \geq 51 \quad (\text{C2})$$

All variables are non-negative.

Find the optimal solution of variable X

Answer 41

Answer range +/- 0.41



Points: 1

11. Calculated Numeric:

BU5401-2024S2::SAE01::Q1101::NUM::(35...



Question Consider the LP Model below:

$$\text{Min } 4X_1 + 30X_2$$

such that

$$-8X_1 + 34X_2 \geq 49 \quad (\text{C1})$$

$$34X_1 + 3X_2 \geq 45 \quad (\text{C2})$$

All variables are non-negative.

Compute the value of X1 at optimal solution

Answer 1.172034

Answer range +/- 0.01172



Points: 1

12. Calculated Numeric:

BU5401-2024S2::SAE01::Q1201::NUM::(38...



Question

In the LP Model below:

$$\text{Min } 25X + 2Y + 26Z$$

such that

$$16X + 9Y + 18Z \geq 119 \quad (\text{C1})$$

$$-6X - 8Y + 21Z \geq 117 \quad (\text{C2})$$

$$25X - 9Y - 4Z \geq 79 \quad (\text{C3})$$

All variables are non-negative.

determine the optimal objective function value

Answer

282.532934

Answer

2.825329

range +/-

Select:

Select by Type:

- Question Type -



Delete

Points

Update

Hide Question Details

Edit Mode is: **ON****BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN**

Tests, Surveys, and Pools

Tests

Test Canvas : SAE 02

Test Canvas: SAE 02 *The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)***Create Question** **Reuse Question** **Upload Questions****Question Settings**

Description Avail: 2025 Jan 21-2300 to Feb 02-2359
(W2 last class to W3 Sun midnight)
Normally till W2 Sun midnight
Max Attempts: 09
Grade: Highest Graded
Password: 752

Instructions

Total Questions 12

Total Points 12

Select: **All** **None**

Select by Type:

- Question Type - **Delete**Points **Update****Hide Question Details**Points: **1****1. Multiple Choice:****BU5401-2024S2::SAE02::Q101::MC::(100,...****Question** In a LP model, the objective function may:**Answer** have quadratic functions of decision variables

not have zero as coefficient

not have infinity as constant term

not have zero as constant term

OK



Points: 1

2. True/False: BU5401-2024S2::SAE02::Q201::TF::
(144,...



Question A under-constrained problem is one in which the objective function can be increased to infinity.

Answer ☒ True
False



Points: 1

3. True/False: BU5401-2024S2::SAE02::Q301::TF::
(169,...



Question It is possible that the optimal solution in a LP model remains the same even when the objective function coefficients are modified slightly.

Answer ☒ True
False



Points: 1

4. Multiple Choice:
BU5401-2024S2::SAE02::Q401::MC::(198,...



Question The 'Allowable Decrease' in the Variable (or Adjustable) Cell section of an Excel LP sensitivity report tells us the extent of downward adjustment of:

Answer ☐ the decision variable's value
☐ the constraint's coefficient
☐ the constraint's right-hand-side value
☒ the decision variable's coefficient



Points: 1

5. Multiple Choice:
BU5401-2024S2::SAE02::Q501::MC::(264,...



Question When a proposed change goes beyond the corresponding Allowable Increase or Decrease in the Adjustable Cells section,

- Answer**
- we conclude that the optimal solution will remain the same.
 - we should conclude that the variable's value will become zero.
 - we conclude that all the shadow prices are invalid.
 - ☒ we would not be able to conclude that optimal solution will not change.



Points: 1

6. Multiple Choice:

BU5401-2024S2::SAE02::Q601::MC::(325,...



Question A concept that is most closely related to the definition of 'shadow price' is:

- Answer**
- bright light
 - distance
 - ☒ gradient
 - reduced cost



Points: 1

7. Multiple Choice:

BU5401-2024S2::SAE02::Q701::MC::(386,...



Question A redundant constraint is ____:

- Answer**
- significantly expands the set of feasible solutions and thereby increases the possibility of an even better optimal solution.
 - ☒ not going to change the optimal solution whether it is present in the model or is removed.

to be removed from the model even if it was the output from a business meeting.

going to significantly reduced the set of feasible solutions.



Points: **1**

8. True/False: BU5401-2024S2::SAE02::Q801::TF::
(446,...



Question Within the bounds of Allowable Increase, increasing the right-hand side of a non-binding constraint will not cause a change in the optimal solution.

Answer ☒ True
False



Points: **1**

9. Multiple Choice:
BU5401-2024S2::SAE02::Q901::MC::(475,...



Question The assumptions necessary for a linear programming model to be appropriate include all of the following except:

Answer ☐ divisibility
☐ knowledge of all constants involved
☒ normality
☐ linearity



Points: **1**

10. Calculated Numeric:
BU5401-2024S2::SAE02::Q1001::NUM::(53...



Question You are given the LP Model as follows:

Min $28X - 10Y + 23Z + 25V$
 such that
 $28X + 12Y + 22Z + 27V \geq 185$ (C1)
 $29X + 4Y + 35Z + 21V \leq 131$ (C2)
 $-4X - 6Y - 12Z + 29V \geq 85$ (C3)
 $-12X + 43Y + 38Z - 5V \leq 182$ (C4)
 All variables are non-negative.

Find the value of variable V at optimal solution

Answer 4.027507

Answer 0.040275
range +/-



Points: 1

11. Calculated Numeric:
BU5401-2024S2::SAE02::Q1101::NUM::(56...



Question Consider the following LP Model:

Min $-4X + 28Y + 29Z$
 such that
 $-9X + 19Y + 40Z \geq 24$ (C1)
 $10X + 5Y + 10Z \leq 135$ (C2)
 $4X + 14Y + 29Z \geq 89$ (C3)
 $9X + 37Y - Z \leq 82$ (C4)
 All variables are non-negative.

Determine the maximum of all non-zero reduced costs (note: sign matters)

Answer 14.171021

Answer 0.14171
range +/-



Points: 1

12. Calculated Numeric:
BU5401-2024S2::SAE02::Q1201::NUM::(59...



Question Examine the following LP Model carefully:

Min $3X_1 + 4X_2 + 16X_3 + 25X_4$
 such that
 $13X_1 + 19X_2 - 14X_3 - 8X_4 \leq 130$ (C1)
 $14X_1 + 12X_2 + 27X_3 - 10X_4 \geq 52$ (C2)
 $32X_1 + 37X_2 + 9X_3 - 7X_4 \leq 156$ (C3)
 $7X_1 + 11X_2 + 6X_3 + 5X_4 \geq 79$ (C4)
 All variables are non-negative.

Calculate the largest non-zero slack or surplus

Answer 92.299311

Answer 0.922993
range +/-

Select: Select by Type:

Delete

Points

Update

Hide Question Details

Edit Mode is: **ON****BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN**

Tests, Surveys, and Pools Tests

Test Canvas : SAE 03

Test Canvas: SAE 03*The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)***Create Question** ▾**Reuse Question** ▾**Upload Questions**

Question Settings

Description Avail: 2025 Jan 28-2300 to Feb 02-2359
(W3 last class to Sun midnight)
Max Attempts: 09
Grade: Highest Graded
Password: 962

Instructions

Total Questions 12

Total Points 12

Select: **All** **None** Select by Type: - Question Type - ▾

Delete

Points

Update

Hide Question Details

Points: **1****1. Multiple Choice:****BU5401-2024S2::SAE03::Q101::MC::(98, ...****Question**

A mutual fund manager must decide how much money to invest in Atlantic Oil (A) and how much to invest in Pacific Oil (P). No more than 40% of the money invested in the two oil companies must be in Pacific Oil. A correct modeling of this constraint is:

← OK

Answer

$$0.6A + 0.4P < 0$$

$$-0.4A - 0.6P < 0$$

$$0.6A - 0.4P < 0$$

☒ $-0.4A + 0.6P < 0$



Points: 1

2. Multiple Choice:

BU5401-2024S2::SAE03::Q201::MC::(151,...



Question

A factory manager wonders how he may optimize his production by tuning the number of workers deployed per day, purchased raw materials (kg) and working hours per day. A decision variable could be:

Answer

hiring agency fee per worker hired

salary rate per worker per hour

cost (\$) per kg of raw materials

☒ number of workers deployed per hour



Points: 1

3. Multiple Choice:

BU5401-2024S2::SAE03::Q301::MC::(205,...



Question

'Ensure that profit-per-week is at least \$5k' is an example of:

Answer

resource enrichment constraint

objective function

☒ performance requirement constraint

resource limitation constraint



Points: 1

4. Multiple Choice:

BU5401-2024S2::SAE03::Q401::MC::(268,...



Question	Chronological order of events is often modeled within LP using:
Answer	☒ numerous decision variables organized by suffixes ☐ time-related units when defining decision variables ☐ defining extra time decision variables as control variables ☐ ordering of constraints with earlier events listed first



Points: 1

5. Multiple Choice:

BU5401-2024S2::SAE03::Q501::MC::(309,...



Question	Profit and revenue are typically modeled as objective functions to be:
Answer	☐ reduced ☐ subsidized ☒ maximized ☐ multiplied



Points: 1

6. Multiple Choice:

BU5401-2024S2::SAE03::Q601::MC::(358,...



Question	Let quantity of production at the end of day i be defined as Q_i for a total modeling period of 3 days. In order to ensure that production quantities at latter days are always greater than earlier days, one way to expressed this could be:
Answer	☒ $Q_1 \leq Q_2, Q_2 \leq Q_3$ ☐ $Q_1 \leq Q_2 \leq Q_3$

$$Q_1 \leq Q_2$$

$$Q_1 \geq Q_2, Q_2 \geq Q_3$$



Points: 1

7. Multiple Choice:

BU5401-2024S2::SAE03::Q701::MC::(407,...



Question

Let S and W be number of Supervisors and Workers deployed at start of a month. If the ratio of Worker-to-Supervisor is at most 5:1, it can be written as:

Answer

☒ $S - 0.2W \geq 0$

$W - S \leq 5$

$S - W \leq 5$

$5W - S \leq 0$



Points: 1

8. Multiple Choice:

BU5401-2024S2::SAE03::Q801::MC::(470,...



Question

A factory makes a product in 3 sequential stages. A conveyor belt carries any product made at previous stage to the next stage at constant speed without spill nor spoilage. Let X_i be the number of products received at stage i and Y_i be the number of products sent out from stage i . The presence of conveyor belt can be modeled as:

Answer

$X_{(i+1)} = Y_{(i+1)}$

$Y_1 = X_3$

$X_i = Y_i$

☒ $Y_i - X_{(i+1)} = 0$



Points: 1

9. True/False: BU5401-2024S2::SAE03::Q901::TF::

(519,...



Question	Non-negativity constraints will be dictated by client or user in real-life.
-----------------	---

Answer	True
	False



Points: **1**

10. Calculated Numeric:

BU5401-2024S2::SAE03::Q1001::NUM::(54...



Question	You are given the LP Model as follows:
-----------------	--

Max $-5X - 4Y + 28Z + 3V$
 such that
 $-5X + 27Y + 42Z + 45V \leq 112$ (C1)
 $-10X + Y - Z + 25V \geq 54$ (C2)
 $21X - 7Y + 41Z + 40V \geq 126$ (C3)
 $-5X - 7Y + 42Z - 4V \leq 117$ (C4)
 All variables are non-negative.

Find the objective function value at optimal solution

Answer	3.412074
---------------	----------

Answer range +/-	0.034121
-------------------------	----------



Points: **1**

11. Calculated Numeric:

BU5401-2024S2::SAE03::Q1101::NUM::(59...



Question

Kerstine is overwhelmed by excitement when she receives the admission for her university studies. She finds it hard to decide how many hours to spend for studying (S), socializing (C), joining hall activities (H), and having personal time (P).

For each category, she associates utility value (in units of U) per hour involved in that category as follows:

	Utility/hr
S	2.8
C	1.6
H	1.8
P	1.1

Removing sleeping, transport and average classroom hours, she only has a net total of 12.0 hrs to accomplish all the above each study day. To stay balanced, she intends to have at least 0.9 hr for socializing, and at least 0.5 hr for personal time. She plans to have at least 1.8 hours of study, 1.7 hours of hall activities, but the total of study and hall activities should be kept within 8.3 hours. Because she also joins study groups which allows her to socialize a bit, so she wants to limit the total hours for study and socializing to no more than 6.1 hours.

What is the minimum daily utility value Kerstine could get out of her university life?

Answer 17.9

Answer range +/- 0.179



Points: **1**

12. Calculated Numeric:

BU5401-2024S2::SAE03::Q1201::NUM::(72...



Question

Alan has a budget of \$1,185 to spend on a vacation and wants to maximize the total duration of relaxation derived from various activities : staying in a hotel, dining at a restaurant, visiting attractions, attending events. He has established a table estimating his duration of relaxation in hours per day for each activity :

Activity	Cost per Day (\$/day)	Relax Hour/Day
Hotel Stay	126	11
Restaurant	56	5
Attractions	38	6
Events	54	7

Alan requires that time spent on events must be at least 22% of the total time spent on all activities.

If Alan achieves maximum total duration of relaxation, how many hours per day will he be getting from Attractions?

Answer 133.569366

Answer range +/- 1.335694

Select:

All

None

Select by Type:

- Question Type -

Delete

Points

Update

Hide Question Details

Edit Mode is: **ON****BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN**

Tests, Surveys, and Pools Tests

Test Canvas : SAE 04

Test Canvas: SAE 04 *The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)***Create Question** **Reuse Question** **Upload Questions****Question Settings**

Description Avail: 2025 Feb 04-2300 to Feb 09-2359
(W4 last class to Sun midnight)
Max Attempts: 09
Grade: Highest Graded
Password: 860

Instructions

Total Questions 12

Total Points 12

Select: **All** **None** Select by Type: - Question Type - 

Delete

Points

Update

Hide Question Details

Points: **1****1. Multiple Choice:****BU5401-2024S2::SAE04::Q101::MC::(97, ...**

Question In an integer linear program, a user wanted to know if constraint #1's RHS was increased by 0.5 unit, what would be the increase in optimal value. The user could:

Answer use shadow price in sensitivity report.

find it impossible to know since all changes must be integer.

solve the ILP twice to calculate difference in optimal values.

OK

never know since sensitivity report is not applicable.



Points: 1

2. Multiple Choice:

BU5401-2024S2::SAE04::Q201::MC::(141,...



Question	In general, integer programming cannot give:
Answer	infinite number of solutions. two solutions. ☒ a close-to-integer real number like 2.00001 no solution.



Points: 1

3. Multiple Choice:

BU5401-2024S2::SAE04::Q301::MC::(189,...



Question	If the acceptance of project A is conditional on the acceptance of project B, and vice versa, the appropriate constraint to use is a:
Answer	k out of n alternatives constraint. multiple-choice constraint. ☒ corequisite constraint. mutually exclusive constraint.



Points: 1

4. Multiple Choice:

BU5401-2024S2::SAE04::Q401::MC::(244,...



Question	Rounding the solution of an LP Relaxation to the nearest integer values provides:
Answer	a feasible but not necessarily optimal integer solution.

☒ corequisite constraint.

an integer solution that is optimal.

an infeasible solution.



Points: 1

5. Multiple Choice:

BU5401-2024S2::SAE04::Q501::MC::(299,...



Question	In an all-integer linear program,
Answer	all constraint coefficients must be integer.
	☒ all solutions must be integer.
	all right-hand side values must be integer.
	all shadow prices must be integer.



Points: 1

6. Multiple Choice:

BU5401-2024S2::SAE04::Q601::MC::(357,...



Question	The solution to the LP Relaxation of a maximization integer linear:
Answer	can be assumed to be the actual optimal solution.
	always forms the basis to generate the actual solution.
	☒ cannot, in general, be rounded off to obtain the actual solution.
	provides the sensitivity report required in the actual ILP model.



Points: 1

7. Multiple Choice:

BU5401-2024S2::SAE04::Q701::MC::(408,...



Question In a model consisting of decision variables X1 and X2, where X2 is required to be integer, the feasible region with X2 being the y-axis would look like:

Answer dots.

☒ horizontal lines on integer y-coordinates.

vertical lines on integer x-coordinates.

vertical lines on any x-coordinate.



Points: 1

8. Multiple Choice:

BU5401-2024S2::SAE04::Q801::MC::(459,...



Question A sensitivity report from Excel for mixed- or all- integer linear programming:

Answer can be provided only by desktop or laptop computer.

☒ cannot be generated.

may be used as an initial estimate.

can be used if any decision variable is integer.



Points: 1

9. Multiple Choice:

BU5401-2024S2::SAE04::Q901::MC::(506,...



Question To perform sensitivity analysis involving an integer linear program, it is recommended to:

Answer use LP relaxation.

☒ make multiple runs of solver for comparisons of effects

use the shadow prices by rounding them to integers first.

use the same approach as you would for a linear program.



Points: 1

10. Calculated Numeric:

BU5401-2024S2::SAE04::Q1001::NUM::(55...



Question

You are given the LP Model as follows:

$$\text{Min } -2X - 6Y - 6Z + 10V$$

such that

$$12X - 12Y + 9Z - 5V \leq 88 \quad (C1)$$

$$10X + 9Y - Z + 37V \geq 38 \quad (C2)$$

$$-X + 34Y + 19Z + 18V \leq 28 \quad (C3)$$

$$X + 44Y + 7V \leq 47 \quad (C4)$$

All variables are non-negative.

Find the objective function value at optimal solution

Answer

-23.143769

Answer

0.231438

range +/-



Points: 1

11. Calculated Numeric:

BU5401-2024S2::SAE04::Q1101::NUM::(59...



Question

Poker cards consisting of all 4 suits of Ace, 2, 3, 4, 5 are laid open on table. Treat Ace as 1. The 4 suits are Spade, Heat, Diamond and Club respectively. Any card chosen will be awarded with the corresponding dollar amount in the reward table. For example, if 5 of Spade (5S) is chosen, then the reward is 11.

Reward \$	Spade ♠	Heart ♥	Diamond ♦	Club ♣
A	100	96	81	89
2	91	76	52	100
3	67	46	22	91
4	36	18	3	67
5	11	1	2	36

Exactly 2 cards must be selected from this deck. The same number in a selected card cannot have more than one suit. Both cards cannot be from the same suit. Spade and Heart cannot be both chosen. Diamond and Club cannot be both chosen.

What is the minimum total that can be obtained?

[NOTE: Just enter the value; do not enter any symbol like "\$" or comma, or decimal point.]

Answer 4

Answer range +/- 0.04



12. Calculated Numeric:

BU5401-2024S2::SAE04::Q1201::NUM::(74...



Points: **1**

Question A large MNC needs to place employees for duty roster at lowest possible total manhour cost for the days M, T, W, Th, F in a week. 4 departments, A, B, C, D require minimum weekly manhours as listed: A=9, B=18, C=10, D=16. Each department pays different per-man hourly rates as follows: A=\$80, B=\$40, C=\$45, D=\$65.

Corporate department demands that each day, total manhour must not exceed the numbers listed below: M=20, T=18, W=17, Th=16, F=14.

What is the total number of staff that will be deployed?

Answer 53

Answer range +/- 0.53

Select:

All

None

Select by Type:

- Question Type -

▼

Delete

Points

Update

Hide Question Details

Edit Mode is: **ON****BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN**

Tests, Surveys, and Pools Tests

Test Canvas : SAE 05

Test Canvas: SAE 05 *The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)***Create Question** **Reuse Question** **Upload Questions**

Question Settings

Description Avail: 2025 Feb 11-2300 to Feb 16-2359
(W5 last class to Sun midnight)
Max Attempts: 09
Grade: Highest Graded
Password: 374

Instructions

Total Questions 12

Total Points 12

Select: **All** **None** Select by Type: - Question Type - 

Delete

Points

Update

Hide Question Details

Points: **1****1. Multiple Choice:****BU5401-2024S2::SAE05::Q101::MC::(97, ...**

Question Which of the following two classes of problems share a great deal of similarities:

Answer transportation and spanning tree.



assignment and transportation.

assignment and network flow.

shortest-path and assignment.

OK



Points: 1

2. Multiple Choice:

BU5401-2024S2::SAE05::Q201::MC::(144,...



Question Which of the following is a commonality between a transportation problem and a shortest-path problem?

Answer destination nodes have non-zero outgoing quantities.



there must be at least 1 source and 1 destination node.

source nodes have non-zero incoming quantities.

both problems require one and only one source node.



Points: 1

3. Multiple Choice:

BU5401-2024S2::SAE05::Q301::MC::(194,...



Question Loops would not form in the shortest-route problem formulated in LP because:

Answer loops add to the total cost and become inferior solutions.

loops will entangle the network and cannot form.

non-negativity constraints in LP ensure loops cannot be formed.

there is only 1 entity at the source.



Points: 1

4. Multiple Choice:

BU5401-2024S2::SAE05::Q401::MC::(239,...



Question Which of the following is not true regarding the typical linear programming formulation of a transportation problem without specialized considerations?

Answer Total destination demands does not exceed total source capacities.

The number of variables is (number of origins) x (number of destinations).

The number of regular constraints is (number of origins) + (number of destinations).

✓ The decision variables are either 0 or 1.



Points: 1

5. Multiple Choice:

BU5401-2024S2::SAE05::Q501::MC::(292,...



Question	The objective of the transportation problem is to minimize:
Answer	Number of origins used to satisfy total demand at the destinations. ✓ Total cost of shipping products from several origins to several destinations. Inconvenience of destinations receiving goods from multiple sources. Number of shipments necessary to satisfy total demand at the destinations.



Points: 1

6. Multiple Choice:

BU5401-2024S2::SAE05::Q601::MC::(341,...



Question	In a typical standard transportation problem, the number of units shipped from origin i to destination j is represented by:
Answer	s_i d_j ✓ x_{ij} c_{ji}



Points: 1

7. Multiple Choice:**BU5401-2024S2::SAE05::Q701::MC::(391,...**

Question In a standard assignment problem with x_{ij} representing assigning agent i to task j , the constraint $x_{31} + x_{32} + x_{33} + x_{34} \leq 2$ means:

Answer ☒ Agent 3 can be assigned up to 2 tasks.

There is no feasible solution.

2 out of tasks 1, 2, 3, 4 may be randomly chosen to be completed by any agent.

Agent 3 must be assigned with 2 tasks.



Points: 1

8. Multiple Choice:**BU5401-2024S2::SAE05::Q801::MC::(443,...**

Question In shortest-path problem, 'total-in equals total-out' applies to:

Answer ☒ Transshipment nodes, source node and destination node.

Bi-directional nodes only.

All nodes other than start and end nodes.

Destination node only.



Points: 1

9. Multiple Choice:**BU5401-2024S2::SAE05::Q901::MC::(499,...**

Question The parts of a network diagram that represent the quantities available at the origins are the:

Answer Demands.

Flows.

Directionalities.

✓ Capacities.



Points: 1

10. Calculated Numeric:
BU5401-2024S2::SAE05::Q1001::NUM::(55...



Question

You are given the LP Model as follows:

$$\text{Min } -3X - 3Y + 23Z + 26V + 27W$$

such that

$$39X - 8Y + 3Z + 29V \geq 68 \quad (\text{C1})$$

$$37X + 13Y - 6Z + 27V + 5W \leq 140 \quad (\text{C2})$$

$$25X + 9Y + 16Z + 17V + 41W \geq 154 \quad (\text{C3})$$

$$42X + 44Y + 13Z + 15V - 11W \geq 46 \quad (\text{C4})$$

$$-X + 38Y + 18Z + 33V + 6W \geq 165 \quad (\text{C5})$$

All variables are non-negative.

Find the optimal objective function value

Answer

31.196601

Answer

0.311966

range +/-



Points: 1

11. Calculated Numeric:
BU5401-2024S2::SAE05::Q1101::NUM::(59...



Question

A logistics company needs to transport discrete units of goods from sources S1, S2 to business units at D1, D2, D3, D4. The sources each have capacities:

S1=725, S2=680.

The destination business units require quantities as follows:

D1=235, D2=105, D3=310, D4=250.

The costs of transporting the goods are given as follows:

Costs \$	D1	D2	D3	D4
S1	52	51	42	43
S2	47	54	33	32

How many goods were delivered from S2 to D3?

Answer 310

Answer range +/- 3.1



Points: **1**

12. Fill in the Blank:

BU5401-2024S2::SAE05::Q1201::FIB::(78...



Question

Berd wishes to travel from Potong Pasir (A) to Tampines (G) at the lowest total cost. The possible intermediate stops, such as bus interchanges, MRT, junctions, bike-sharing stations, etc are represented as letters. The table below gives the traveling fees when traveling from "From Location" to "To Location".

From Node	To Node	Fees \$
A	C	47
B	E	79
C	B	72
C	E	62
D	A	21
D	B	55
D	C	36
D	E	84
D	F	29
D	G	54
E	A	42
E	D	93
E	F	79
F	A	69
F	B	72
G	B	95
G	C	79
G	D	79
G	E	67
G	F	48

Routes that are not listed are not usable.

What is the shortest path for Berd?

NOTE: If X-Y-Z is your answer, enter XYZ in exactly the same casing and letter as the given cost table. Do not include any dash "-", space or letters not used in the cost table.

Evaluation Method	Answer	Case Sensitivity
Exact Match	ACEDG	

Select:
 Select by Type:

--



Edit Mode is: ☒ ON



BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN

Tests, Surveys, and Pools

Tests

Test Canvas : SAE 06

Test Canvas: SAE 06

The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)

Create Question 

Reuse Question 

Upload Questions



Question Settings

Description Avail: 2026 Feb 18-2300 to Feb 23-2359
(W6 last class to Sun midnight)
Max Attempts: 09
Grade: Highest Graded
Password: 502

Instructions

Total Questions 12

Total Points 12

Select: Select by Type: 



Delete

Points

Update

Hide Question Details

← OK



Points: 1

1. Multiple Choice:**BU5401-2024S2::SAE06::Q101::MC::(97, ...**

Question	Simulation:
Answer	☒ All of the alternatives are correct. does not guarantee optimality. allows testing of the system without affecting the real system. is flexible and does not require the assumptions of theoretical models.



Points: 1

2. Multiple Choice:**BU5401-2024S2::SAE06::Q201::MC::(143,...**

Question	A quantity that is difficult to measure with certainty is called a:
Answer	☐ risk-adjusted measurement. risk analysis. ☒ probabilistic input. profit/loss process.



Points: 1

3. Multiple Choice:**BU5401-2024S2::SAE06::Q301::MC::(190,...**

Question	The best way to describe Excel's RANDBETWEEN(a,b) function is:
Answer	☐ discrete uniform distribution excluding all values between a to b. continuous uniform distribution from a to b.

non-probabilistic distribution.

- ☒ discrete uniform distribution of all integers between a and b.



Points: 1

4. Multiple Choice:

BU5401-2024S2::SAE06::Q401::MC::(238,...



Question

Which of the following is not true about cumulative distribution function $cdf(x)$ of random variable X :

Answer

Its curve may start from -infinity and may end at infinity.

- ☒ Its concept only defined for continuous distribution.

It is always between 0 to 1 inclusive.

It is strictly non-decreasing.



Points: 1

5. Multiple Choice:

BU5401-2024S2::SAE06::Q501::MC::(292,...



Question

For a continuous distribution used in a simulation, the cumulative distribution function $cdf(x)$ is related to the inverse cdf function $invCdf(p)$ by:

Answer

- ☒ $invCdf(cdf(x)) = x$

$x * cdf(x) = p * invCdf(p)$

$x / cdf(x) = p / invCdf(p)$

$cdf(x) * invCdf(p) = 1$



Points: 1

6. Multiple Choice:

BU5401-2024S2::SAE06::Q601::MC::(347,...



Question	The word 'uniform' in the term 'uniform random numbers' means:
Answer	All the numbers are odd or all are even. All numbers are uniformly spaced apart from each other. All the numbers have the same number of digits. ☒ Each number has an equal probability of being drawn.



Points: 1

7. Multiple Choice:

BU5401-2024S2::SAE06::Q701::MC::(399,...



Question	Flight simulator is an example of simulation in which the systemic behavior of the system is the successful taking off and landing of the plane. Which of the following is not a plausible component of the flight simulation system?
Answer	☒ Mistakes made by pilot. Engine failure and resulting plane behavior. Direction and intensity of sunlight. Throttle.



Points: 1

8. Multiple Choice:

BU5401-2024S2::SAE06::Q801::MC::(453,...



Question	Which of the following is not a suitable use of simulation:
-----------------	---

Answer



Solution of simultaneous equations.

Shooting games.

Weather forecast.

Drug design.



Points: 1

9. True/False: BU5401-2024S2::SAE06::Q901::TF::
(501,...



Question

One disadvantage of simulation is that it is limited in the variety of probability distributions that can be used in modeling a system.

Answer

True



False



Points: 1

10. Calculated Numeric:
BU5401-2024S2::SAE06::Q1001::NUM::(52...



Question

Let X be a random variable which follows normal distribution with mean 520 and standard deviation 161.2.
Calculate cdf(776) of X.

Answer

0.943867

Answer

0.009439

range +/-



Points: 1

11. Calculated Numeric:
BU5401-2024S2::SAE06::Q1101::NUM::(59...



Question

Suppose X is a random variable which follows uniform distribution from 560 to 794.
Calculate invcdf(0.2925) of X.

Answer

628.445

Answer

6.28445

range +/-

Points: 1



12. Calculated Numeric:

BU5401-2024S2::SAE06::Q1201::NUM::(65...



Question

X is a random variable which has a exponential distribution with a mean of 0.5. Using the following table of random values (as generated from Excel's RAND()), calculate the sum of the simulated values of X.

Random Values
0.5225
0.2538
0.5974
0.2624
0.2267

Answer 1.251603

Answer range +/- 0.012516



Select:

Select by Type:

- Question Type -

Delete

Points

Update

Hide Question Details



Edit Mode is: ON



BU5401-MANAGEMENT DECISION TOOLS (SEM) AY24/25 SEM 2 MAIN

Tests, Surveys, and Pools

Tests

Test Canvas : SAE 07

Test Canvas: SAE 07

The Test Canvas lets you add, edit, and reorder questions, as well as review a test. [More Help](#)

Create Question

Reuse Question

Upload Questions



Question Settings

Description Avail: 2025 Feb 25-2300 to Mar 02-2359
(W7 last class to Sun midnight)
Max Attempts: 09
Grade: Highest Graded
Password: 296

Instructions

Total Questions 12

Total Points 12

Select: All None Select by Type: - Question Type -



Delete

Points

Update

Hide Question Details



Points: 1

1. Multiple Choice:

BU5401-2024S2::SAE07::Q101::MC::(97, ...



Question Forecasting may be grouped into:

Answer Accurate and inaccurate forecasting.



Short-, mid- and long-term forecasting

Approximation-oriented and deterministic forecasting.

Computer-based and human-based forecasting.

← OK

Previous



2. True/False: BU5401-2024S2::SAE07::Q201::TF::

(146,...



Question Moving average with large window size will be less responsive to transient changes in values.

Answer ☒ True
False



Points: **1**

3. True/False: BU5401-2024S2::SAE07::Q301::TF::

(169,...



Question A four-period moving average forecast for period 10 would be found by averaging the values from periods 10, 9, 8, and 7.

Answer True
☒ False



Points: **1**

4. Multiple Choice:

BU5401-2024S2::SAE07::Q401::MC::(196,...



Question Which of the following forecasting methods puts the most weight on the most recent time series value?

Answer exponential smoothing with $\alpha = 0.2$

☒ moving average using the most recent 3 periods

exponential smoothing with $\alpha = 0.3$

moving average using the most recent 4 periods.



Points: **1**

5. Multiple Choice:

BU5401-2024S2::SAE07::Q501::MC::(241,...



Question Seasonal patterns:

Answer are multi-year runs of observations above or below the trend line.

reflect a shift in the time series over time.

cannot be predicted.

✔ are regular repeated patterns.



Points: 1

6. Multiple Choice:

BU5401-2024S2::SAE07::Q601::MC::(289,...



Question

Which of the following exponential smoothing constant values puts the same weight on the most recent time series value as does a 5-period moving average?

Answer

✔ alpha = 0.2

alpha = 0.1

alpha = 0.8

alpha = 0.75



Points: 1

7. True/False: BU5401-2024S2::SAE07::Q701::TF::

(334,...



Question

Moving average methods work on the assumption that any data value above the prevailing average would be averaged down by future data values below the prevailing average.

Answer

✔ True

False



Points: 1

8. Multiple Choice:

BU5401-2024S2::SAE07::Q801::MC::(361,...



Question

The word 'smoothing' in smoothing methods essentially means:

Answer

Ensuring that cyclical fluctuations are smoothed out.

Reducing the effects of trend.

☒ Averaging out the noises.

Countering effects of seasonality.



Points: 1

9. Multiple Choice:

BU5401-2024S2::SAE07::Q901::MC::(409,...



Question

The MSE for Forecast A is 373,216 while MAE for Forecast B produces 648.7. We can conclude that:

Answer

☒ no conclusion can be made about whether A or B is more accurate.

Both A and B are accurate because the values are at least 100.

B is more accurate than A.

Both A and B are bad because the values have exceeded 10.



Points: 1

10. Multiple Choice:

BU5401-2024S2::SAE07::Q1001::MC::(473...



Question

Which of the following is not an example of time-series forecasting:

Answer

ES-With-Trend method.

☒ Causal forecasting

Moving average.

Exponential smoothing.



Points: 1

11. Calculated Numeric:

BU5401-2024S2::SAE07::Q1101::NUM::(53...



Question

Examine the following daily profit stream:

Day	Daily Profits
1	153
2	289
3	175
4	229
5	372
6	274
7	310
8	398
9	435
10	296

forecast the next daily profit using a moving average with a window of 2 days.

Answer

365.5

Answer

3.655

range +/-



12. Calculated Numeric:

BU5401-2024S2::SAE07::Q1201::NUM::(59...



Points: **1**

Question Consider the following monthly revenue records:

Month	Monthly Revenue
1	1711.2
2	1378.2
3	4592.6
4	1186.2
5	1115.3
6	1490.2
7	1143.6
8	1639.9
9	1698.9
10	3232.9

Using exponential smoothing forecasting method with an alpha of 0.84 and an initial forecast of 1,117, forecast the next monthly revenue.

Answer 2,984.113561

Answer 29.841136
range +/-

Select: Select by Type:

Points